

Financial Management and Valuation

Unit 3: Mini Cases

SBWL Finance

Winter Semester 2025

Please upload your solution and indicate which Mini-Cases (MCs) you have solved before **January 6, 2026 at 10pm**. To do so, go to the assignment section on Canvas and choose “**Unit 3 Indication and Submission of MCs**”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). **There will be no deadline extensions! Only complete submissions containing answers to the T/F questions and a solutions file upload will be considered, your last complete attempt counts.**

You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solution consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Please make sure to hand in only one file of each file type (you might have to merge your scans to one pdf file). Uploaded files will be checked for plagiarism.

The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini-Cases you feel confident to present.

Exercise 3.1: You hold an American call option with a \$20 strike price on a stock that is currently priced at \$25. The option expires in one year. Compare the cash flows you would have at expiration under the following two choices:

- (a) Exercise the call immediately, collect the \$5 intrinsic value, and place this amount in a bank account until the option's expiration date.
- (b) Short-sell the stock today, deposit the \$25 proceeds in a bank account, and keep the call option until expiration.

Exercise 3.2:

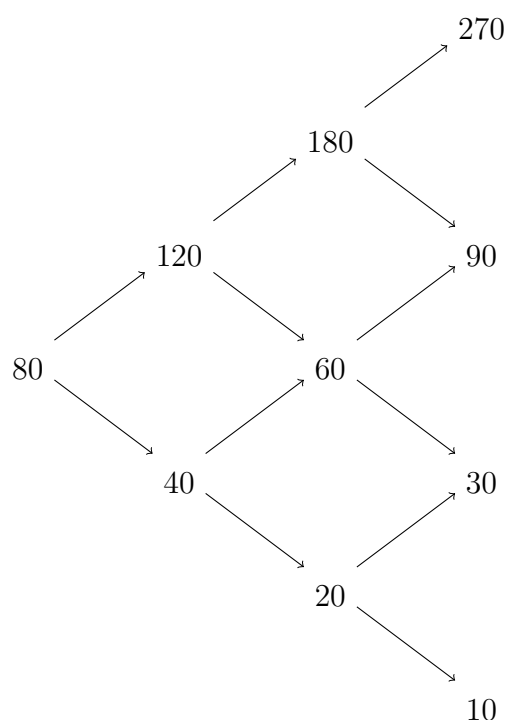
The current price of an equity share of Nvidia (NVDA) is 150. The equity follows a binomial process where each period the share price either goes up 11% or down 7%.

- (a) Draw the two-step tree to value a put option.
- (b) Compute the market value of an European put option on NVDA equity with a strike price of 155 and two periods to expiration. Assume NVDA pays no dividends over the next two periods. The risk-free rate is 3.5% per period (compounded per period).
- (c) Repeat the same exercise for an American put option on NVDA equity. What differs?

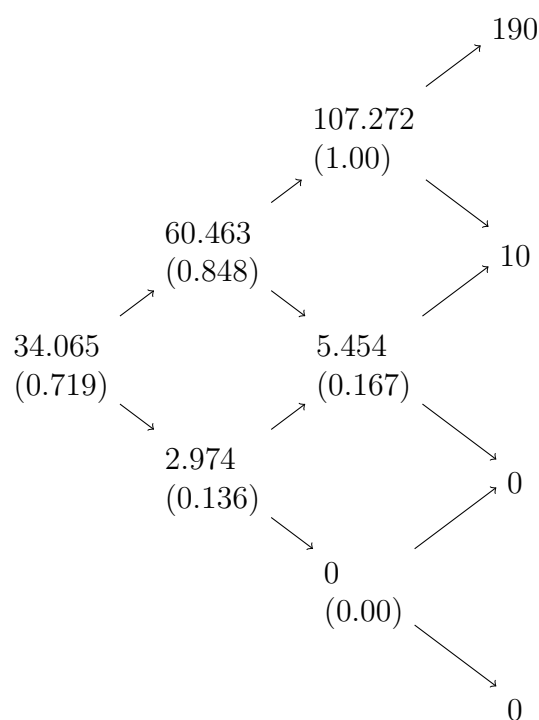
Exercise 3.3:

The first tree below gives the paths the price of a non-dividend paying stock may follow. The second tree displays values of an American call option on this stock and the delta (Δ) of the tracking portfolio in brackets. The two trees are based on the following values: $S = 80$, $N = 3$ (number of binomial periods), $K = 80$, $u = 1.5$, $d = 0.5$ and a per period risk-free return of 10%. Suppose that in period $N = 0$ you sell the call option at the price of 34.065.

Paths of price of underlying:



Paths of call price and Δ of tracking portfolio:



- (a) Hedge your position in the option with the tracking portfolio. Describe how you need to adjust the tracking portfolio node-by-node, starting at period $N = 0$. Show that at option expiry in period $N = 3$ you are hedged. Do so for the following two paths: up-down-down and up-down-up.
- (b) Suppose the option holder exercises too early. Could this behaviour harm your hedging strategy?

Exercise 3.4: An investor buys a European put on a share for \$3. The stock price is \$42 and the strike price is \$40. Under what circumstances does the investor make a profit? Under what circumstances will the option be exercised? Draw a diagram showing the variation of the investor's profit with the stock price at the maturity of the option

Exercise 3.5: You are analysing the capital structure of a privately owned company. Based on your estimates, the current market value of the firm's assets is €52 million, and the annual volatility of the asset return is approximately 0.30. The firm has outstanding zero-coupon bank debt with a promised repayment of €45 million in one year. The annual risk-free interest rate is 3% (annual compounding). Using a one-period binomial model, determine the market value of the firm's debt. Interpret risky debt as a portfolio of risk-free debt and a short put on the assets of a firm, with a strike price equal to the face value of debt.

- (i) Treat the firm's assets as the underlying security. Compute the up and down factors and draw the corresponding one-period binomial tree. What are the put option

payoffs at maturity?

- (ii) Using the risk-neutral probability from the binomial model, determine the present value of the put option.
- (iii) Calculate the market value of risky debt.

Exercise 3.6

You are employed as a junior quantitative analyst at an investment bank. To control the bank's exposure to movements in the U.S. technology sector, you are asked to determine the fair value of a call option written on a major U.S. equity benchmark. The underlying asset is the **NASDAQ-100** index, which tracks 100 of the largest non-financial companies traded on the Nasdaq exchange.

Download one year of *daily closing prices* for the NASDAQ-100 from: <https://www.nasdaq.com/market-activity/index/ndx/historical>.

- (a) Using the historical data, estimate the potential index values one year from now within a one-period binomial framework. Proceed as follows:
 - (i) Compute the series of daily log-returns:

$$r_t = \ln(P_t) - \ln(P_{t-1}).$$
 - (ii) Calculate the sample variance of the daily log-returns.
 - (iii) Compute the annualized variance using the number of trading days in a year, and obtain the annualized standard deviation.
 - (iv) With a single binomial step ($N = 1$) and a one-year horizon ($T = 1$), compute the up and down factors, u and d .
 - (v) Using the current index level, compute the possible “up” and “down” index values in one year.
- (b) Assume that the annual risk-free rate is 2.5% (annually compounded). Compute the risk-neutral probability of an up movement.
- (c) Using your results, determine the arbitrage-free price of a European call option with strike equal to today's index level.